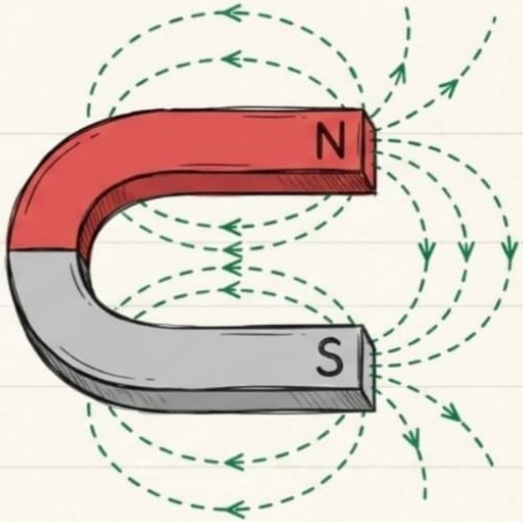
Magnetic Field & Magnetism

Comprehensive Study Notes



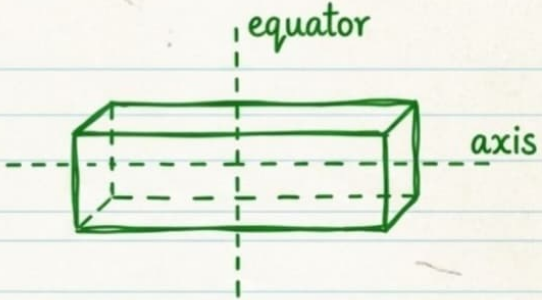
Magnetic Dimensions & Coulomb's Law

* **Magnetic axis**: The line passing through the poles of a magnet is called as magnetic axis.

* **Magnetic equator**: The line passing through the center of the magnet & at ⊥ (perpendicular) to the axis.

* **Magnetic length**: The distance b/w 2 poles of a magnet is called the magnetic length of the magnet.

$$\frac{\text{Magnetic length}}{\text{Geometric length}} = 0.84$$



Coulomb's Law in magnetism

$$F \propto q_{m1} \cdot q_{m2}$$

$$F \propto \frac{1}{r^2}$$

$$\boxed{\therefore F = \frac{\mu_0}{4\pi} \cdot \frac{q_{m1} \cdot q_{m2}}{r^2}}$$

* q_m represents pole strength!

Magnetic Dipole & Moment

Magnetic Dipole: It is similar to the electric dipole. It is the arrangement of 2 equal & opposite magnetic poles separated by a distance 2ℓ.

Magnetic Dipole Moment: It is defined as the product of pole strength & magnetic length. Its direction is from S → N. Its SI unit is $A \cdot m^2$. SI unit of pole strength is $A \cdot m$.

$$\boxed{\vec{m} = q_m \times 2\vec{\ell}}$$

Q: A magnetic dipole has length 10 cm & pole strength 20 A·m. Calculate its magnetic dipole moment.

$$\vec{m} = q_m \times 2\vec{\ell}$$
$$= 20 \times 10 \times 10^{-2}$$
$$= 2\ A \cdot m^2$$

Torque & Gauss's Law Introduction

$$\vec{\tau} = \vec{m} \times \vec{B} = mB\sin\theta$$

Calculation for 180°:

$$= \left(\frac{6 \times 44}{100}\right) \times \sin 180°$$

$$= \frac{6 \times 44 \times 0}{100} = 0\,N$$

Angle	120°	180°
Sin	$\frac{\sqrt{3}}{2}$	0
Cos	$-\frac{1}{2}$	-1
Tan	$-\sqrt{3}$	0

Q: State Gauss's law in magnetism & what are its important consequences?

By Gauss's law in electrostatics,

$$\oint \vec{E} \cdot d\vec{s} = \frac{q_{encl}}{\varepsilon_0}$$

Now, to find flux for an electric field dipole. When this dipole is enclosed in a gaussian surface, then $q_{encl} = 0$.

$$\therefore \oint \vec{E} \cdot d\vec{s} = 0$$

Gauss's Law in Magnetism

Now, if taken a magnet which always stays as a dipole, So,

$$\boxed{\oint \vec{B} \cdot d\vec{s} = 0}$$

Important consequences:

1. This indicates that there are no sources or sinks for B due to a dipole in a closed surface.
2. Hence isolated magnetic poles don't exist.
3. Magnetic poles always exist as unlike poles.
4. The no. of magnetic lines of force entering a closed gaussian surface is equal to no. of magnetic lines leaving the surface.

Some Important terms used to describe magnetic properties:

Magnetising field: A magnetic field that exists in vacuum & induces magnetism in a magnetic material placed inside it with a field i.e., $B = \mu_0 nI$

Core Magnetic Parameters

Magnetic Induction

Defined as total no. of magnetic field lines passing normally per unit area through a material.

↳ SI unit: $T / Wb \cdot m^{-2}$

$$\phi = \vec{B} \cdot \vec{A} \quad \text{or} \quad \frac{\phi}{A} = B$$

Magnetising Field Intensity (H)

Defined as ability of a magnetising field to magnetise a material medium. Denoted by $\vec{H}$, given by no. of ampere turns per unit length of solenoid.

↳ $\vec{H} = n \cdot I$

Intensity of Magnetisation (M)

Defined as magnetic moment developed per unit volume of a substance.

↳ $\vec{M} = \frac{\vec{m}}{V}$, where $m = I \times A = nIA \sin\theta$

Permeability & Susceptibility

Magnetic permeability (μ): Measure of extent to which material can be penetrated by magnetic field. $\mu = \frac{B}{H}$

Relative permeability (μ_r): $\mu_r = \frac{\mu}{\mu_0}$

Magnetic susceptibility (χ_m): Ability of substance to take up magnetisation. $\chi_m = \frac{M}{H}$

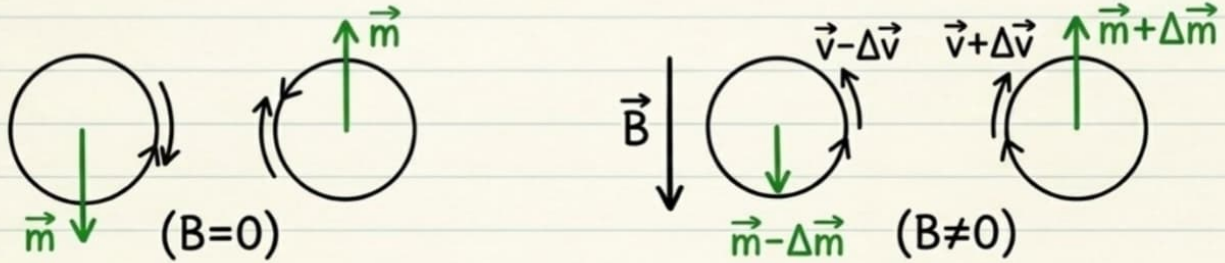
Faraday's Classification of Substances

On the basis of behaviour in external magnetic field, Faraday classified materials into 3 categories:

Diamagnetic substances	Paramagnetic substances	Ferromagnetic substances
- Develop feeble magnetisation magnetisation in the opposite direction of magnetising field. - Repelled by magnets & move from strong field to weak field (away from poles).	- Develop feeble magnetisation in direction of field. - Attracted by magnets & move from weak field to strong field (towards poles).	- Develop strong magnetisation in direction of field. - Strongly attracted by magnets & move from weak field to strong field (towards poles).
- E.g.: Bi, Cu, Pb, Zn, Au etc.	- E.g.: Al, Cr, Pt, Na etc.	- E.g.: Fe, Ni, Co etc.

Origin of Diamagnetic Substances

In a diamagnetic material, the magnetic moment due to different e^- cancel out. One e^- of the pair revolves in clockwise & the other one anti-clockwise which results in 0 net magnetic moment of an atom.

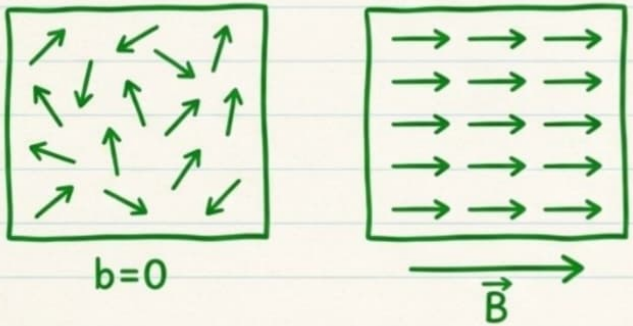


When such a diamagnetic atom is brought near a magnetising field, this will affect the speed of revolution... Revolving velocity decreases away from center and increases towards it. As v decreases, so t increases & thus, I increases along with $\vec{m}$. Since $\vec{m}$ is opposite to magnetising field $\vec{B}$, net $\vec{m}$ will lead to repulsion of the material from magnet.

Diamagnetic materials are independent of each other.

Origin of Paramagnetic Substances & Curie's Law

The atoms of a paramagnetic material possess a permanent magnetic moment due to presence of unpaired e^- or due to non-cancellation of spins.



In presence of external magnetising field, magnetic moment develops in the direction of the magnetising field... thus called as paramagnetic substances.

Curie's Law

This law states that intensity of magnetisation is directly proportional to magnetising field intensity & inversely proportional to temp.

$$\therefore M \propto H$$

$$M \propto \frac{1}{T}$$

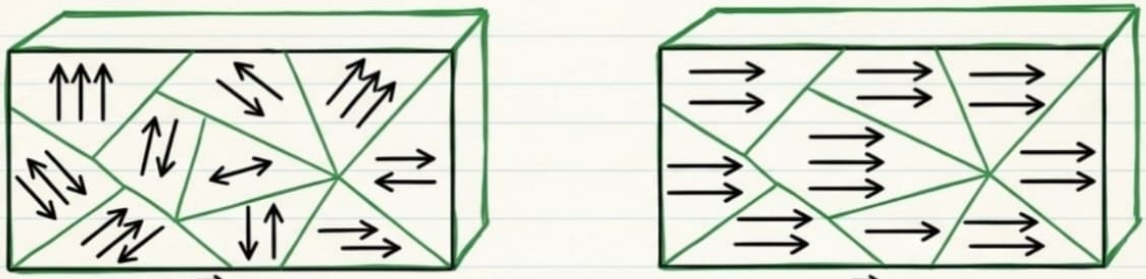
$$\therefore M \propto \frac{H}{T} \quad \text{or} \quad M = C \cdot \frac{H}{T}$$

(where C is Curie's constant).

$$\boxed{\therefore \frac{M}{H} = \frac{C}{T} = \chi_m}$$

Origin of Ferromagnetic Substances

Weiss explained ferromagnetic substances on the basis of domains theory. In materials like Fe, Ni, Co etc., individual atoms are associated with large magnetic moments & interact with neighbour atoms to give same magnetic moment over a region of space known as domains.



$\vec{B} = 0$ $\qquad$ $\vec{B} \neq 0$

Modified Curie's Law:

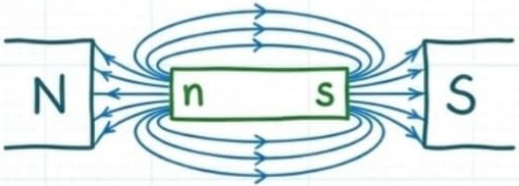
When a ferromagnetic sample is heated, its magnetisation decreases. On sufficient high temperature, it converts into a paramagnetic material. The temperature at which it converts is called Curie point or Curie temperature (T_c).

$$\boxed{\chi_m = \frac{C'}{T - T_c}}$$

(where C' is modified Curie constant. This is Curie-Weiss law).

Properties: Diamagnetic Substances

1. Develops feeble magnetisation in the opposite direction of the applied field.
2. Poles are induced in it in a direction opposite to that of inducing field.
3. Lines of forces prefer to pass through the surrounding air than to pass through the material itself. i.e., lines get expelled or repelled.

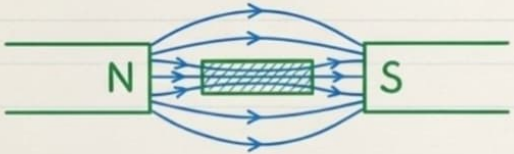


4. Moves from stronger to weaker parts of non-uniform field.
5. Suspended freely in uniform field, it aligns perpendicularly to the magnetising field.
6. Susceptibility $\left(\chi_m = \frac{M}{H}\right)$ is small and −ve. For Bi, $\chi_m = -0.00015$.
7. Relative permeability μ_r $(= 1 + \chi_m)$ is +ve but < 1.
8. Susceptibility lasts so long as the magnetising field is applied.

Properties: Paramagnetic Substances

1. Develops feeble magnetisation in the direction of the applied field.

2. Lines of force prefer to pass through it than through surrounding air i.e., get slightly more conc. inside material.



3. Moves from weaker to stronger parts of non-uniform field.

4. Suspended freely, aligns parallel to magnetising field.

5. Susceptibility has small but +ve value. For Al, $\chi_m = 1{\cdot}8 \times 10^{-6}$.

6. Relative permeability ($\mu_r = 1 + \chi_m$) has a value slightly > 1.

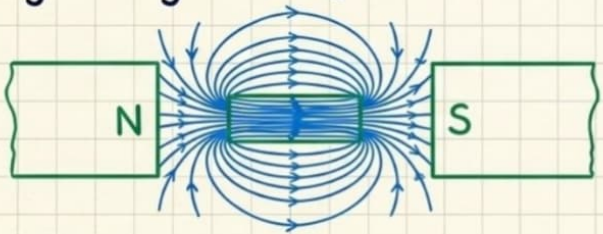
7. Susceptibility varies inversely as absolute temp ($\chi_m \propto \frac{1}{T}$, Curie's law).

8. Intensity of magnetisation is proportional to field; susceptibility/permeability do not vary with field B_0.

9. Loses magnetism as soon as field is removed.

Properties: Ferromagnetic Substances

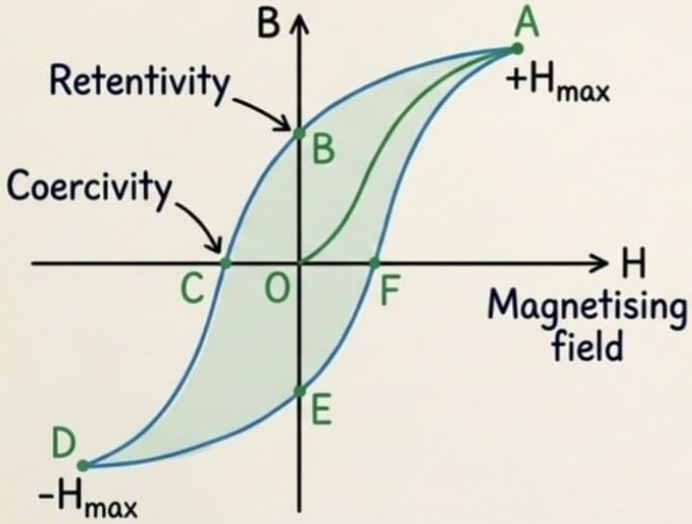
1. Develops strong magnetisation in the direction of the applied field.
2. Lines of force concentrate greatly into the material so that magnetic induction B becomes much more than magnetising field B_0.



3. Moves from weaker to stronger parts of non-uniform field.
4. Suspended freely, quickly aligns parallel to magnetic field.
5. Intensity of magnetisation M is proportional to H for smaller values.
6. Susceptibility $\left(\chi_m = \frac{M}{H}\right)$ has a large +ve value. $(\because M \gg H)$.
7. Relative permeability (μ_r) has large +ve value. (e.g., Fe has $\mu_r = 1000$).
8. Susceptibility decreases with temp in accordance with Curie-Weiss law $\left(\chi_m = \frac{C'}{T - T_c}\right)$.
9. At Curie point, susceptibility suddenly falls & becomes paramagnetic.

The Hysteresis Loop

The phenomenon of lagging of magnetic induction behind magnetising field is called hysteresis (Greek word meaning: delayed).



1. Point O represents initial unmagnetised state.
2. As H increases, B gradually increases, saturating at A ($+H_{max}$).
3. As H decreases to 0, B decreases along path AB. B doesn't become 0. The induction OB left behind is Retentivity (residual magnetism).
4. To reduce magnetisation to 0, H is increased in opposite direction to point OC. This required reverse field is called Coercivity.

Hysteresis Completion & Practical Applications

5. On further increasing H in opposite direction to $-H_{max}$, we reach saturation at D. If H is decreased, it ultimately reaches back to pt A via DEFA.

   The closed curve ABCDEFA is called the hysteresis loop. 'Throughout the cycle, it is clearly visible that $\vec{B}$ always lags behind H'.

Practical importance: From a loop, we can judge parameters like retentivity & coercivity. This helps in selecting proper cores of a transformer and electromagnets for making permanent magnets.

Types of Ferromagnetic Materials

Soft Ferromagnet

- Magnetisation almost disappears after removal of ext field.
- Narrow HL (Hysteresis Loop). Low retentivity, coercivity & HL area.
- Higher μ_r. Used in transformer cores & solenoids.
- E.g.: soft iron, stalloy.

Hard Ferromagnet

- Retentivity of magnetisation even after removal of ext field.
- Wide HL. High retentivity, coercivity & HL area.
- Lower μ_r. Used in making permanent magnets.
- E.g.: steel, alnico, load stone.